

Chasing π : Why the Decimal Never Terminates

Predicting π from Inverse Partition Mechanics of a 512-Bit Spherical Register

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Abstract

Using only the fixed inverse cascade $\div 2 \div 4 \div 4 \div 4 \div 4 = \div 512$ on a closed dual sphere, we demand that both a true 90° arc and a true $\pi \times 90$ arc land on integer bits. The unique minimal dual sphere that satisfies both conditions is exactly 720° . This mechanically forces a 512-bit closed register in which $90^\circ = 64$ bits exactly and $\pi \times 90 \rightarrow 201$ bits exactly. Thus the register's internal constant is $\pi_{\text{register}} = 201/64 = 3.140625$ (exact, dyadic, terminating). The standard irrational $\pi \approx 3.1415926535\dots$ appears only when this finite quaternary/bit-based structure is measured with an infinite-resolution base-10 ruler. Inside the register the chase is over.

All Numbers Are Mechanically Forced — Nothing Is Assumed

□ Fixed law: one dual split $\div 2$ + four quaternary inversions $\div 4^4 \rightarrow \div 512$ total.

Two demands only:

- 90° occupies an integer number of bits ($Q \in \mathbb{Z}$)
- $\pi \times 90$ lands within < 0.5 bit of an integer address ($P \in \mathbb{Z}$ after rounding)

The unique minimal dual extent yielding both is $\Theta = 720 \rightarrow Q = 64$ exactly, $P = \text{round}(\pi \times 64) = 201$ exactly.

No smaller integer-degree sphere works.

1 The 512-Bit Closed Register — Complete Mechanical Derivation

Step	Result (forced)
Dual sphere extent	720°
Inverse cascade	$\div 512$
Register size	512 bits
Bit angle	$720/512 = 45/32 = 1.40625$ per bit
90° arc	exactly 64 bits ($Q = 64$)
$\pi \times 90$ inverse-projected	$\pi \times 64 \approx 201.06193\dots \rightarrow 201$ bits ($P = 201$)
Register π	$\pi_{\text{register}} = 201/64 = 3.140625$ exactly
Structural offset	$201 - 64 = 137$

Why 720°?

The inverse cascade $\div 512$ is fixed. We need:

1. $Q = \Theta / (8 \times \Delta)$ must be an integer (where $\Delta = \Theta / 512$)
2. This simplifies to: Θ must be divisible by 8, so $\Theta = 8k$ for some integer k
3. We also need $P = \pi \times Q$ to land near an integer
4. Testing: $\Theta = 360$ gives $Q = 32$, $P \approx 100.53$ (not close enough)
5. Testing: $\Theta = 720$ gives $Q = 64$, $P \approx 201.06$ (rounds to 201 exactly)

720° is the minimal dual sphere that satisfies both integer constraints.

The Mechanical Forcing

$$\begin{aligned} \text{Dual sphere: } \Theta &= 720 \\ \text{Cascade: } N &= 512 \text{ bits} \\ \text{Bit angle: } \Delta &= \frac{720}{512} = 1.40625 \\ \text{Quadrant: } Q &= \frac{N}{8} = 64 \text{ bits} \\ \text{Quadrant angle: } Q \times \Delta &= 64 \times 1.40625 = 90 \quad \checkmark \\ \text{Pi position: } P &= \text{round}(\pi \times 64) = 201 \text{ bits} \\ \text{Pi angle: } P \times \Delta &= 201 \times 1.40625 = 282.65625 \\ \text{Register constant: } \pi_{\text{register}} &= \frac{P}{Q} = \frac{201}{64} = 3.140625 \end{aligned}$$

Closure proof:

$$N \times \Delta = 512 \times 1.40625 = 720 \quad (\text{zero remainder})$$

2 Octave Scaling — The Ratio Never Changes

Each octave multiplies address density by 4 (one additional quaternary refinement stage). For octave n :

$$N_n = 512 \times 4^n, \quad Q_n = 64 \times 4^n, \quad P_n = 201 \times 4^n$$

The ratio remains invariant:

$$\frac{P_n}{Q_n} = \frac{201 \times 4^n}{64 \times 4^n} = \frac{201}{64} = 3.140625 \quad \forall n \in \mathbb{N}$$

n	Total bits	Q_n	P_n	$\pi_n = P_n/Q_n$	Offset $P_n - Q_n$
0	512	64	201	3.140625	137
1	2,048	256	804	3.140625	548
2	8,192	1,024	3,216	3.140625	2,192
3	32,768	4,096	12,864	3.140625	8,768
4	131,072	16,384	51,456	3.140625	35,072
5	524,288	65,536	205,824	3.140625	140,288
6	2,097,152	262,144	823,296	3.140625	561,152
7	8,388,608	1,048,576	3,293,184	3.140625	2,244,608
8	33,554,432	4,194,304	13,172,736	3.140625	8,978,432

At every finite octave, $\pi = 3.140625$ exactly and terminates.

The offset $P_n - Q_n = 137 \times 4^n$ also scales by powers of 4, maintaining the structural relationship.

3 Two Domains, Two Constants

Structure Domain (Closed 512-Bit Register, Base-4)

Inside the register:

$$\pi_{\text{register}} = \frac{201}{64} = 3.140625 \quad (\text{exact, finite, closed})$$

This is:

- A dyadic rational ($64 = 2^6$)
- Exact in binary: 11.001001_2
- Terminating in decimal: 3.140625 (6 digits)
- The register's native 90-normalized projection constant
- Invariant across all octaves

Projection Domain (Infinite Linear Shadow, Base-10)

When measured with continuous coordinates:

$$\pi_{\text{standard}} \approx 3.1415926535 \dots \quad (\text{irrational, non-terminating})$$

This is:

- The continuum circle constant
- Irrational (not expressible as a ratio of integers)
- Transcendental (not the root of any polynomial with rational coefficients)
- Non-terminating in any integer base
- The infinite-resolution projection of the finite register

The Relationship

The infinite decimal π_{standard} is the artifact of projecting a closed quaternary/bit lattice (base-4 structure with dual-rail binary foundation) into an incommensurable base-10 measurement framework.

The register operates in its native base (quaternary with binary rails). When you impose a foreign measurement system (base-10 decimal coordinates), you get an infinite non-repeating expansion because the bases are incommensurable.

Cross-domain signature (not an error):

$$\sigma_{\pi} = \frac{\pi_{\text{register}} - \pi_{\text{standard}}}{\pi_{\text{standard}}} \times 100\% \approx -0.0308\%$$

This is a stable relation between two different operational domains, not an error in either domain.

4 Why the Decimal Never Terminates

Inside the Register

The register structure is:

- Closed at 512 bits (256/256 dual structure)
- Based on quaternary inversions ($\div 4^4$)
- Dual-rail binary at foundation
- $\pi = 201/64$ exactly (dyadic rational)
- All positions are exact integers
- No infinite sequences needed

The Linear Projection

When you measure this structure with a base-10 ruler:

1. The ruler has infinite resolution (can always subdivide further)
2. Base-4 (quaternary) and base-10 (decimal) are incommensurable
3. Dyadic rationals like $201/64 = 201/2^6$ have terminating binary representations
4. But $201/64$ in base-10 requires: $201 \div 64 = 3.140625$
5. This terminates only because $64 = 2^6$ and $10 = 2 \times 5$
6. For the actual $\pi \approx 3.1415926535\dots$, no termination occurs in any integer base

The sphere closed once. The decimal chase is the shadow of measuring a closed structure with infinite subdivision.

Computing Trillion-Digit Records

When mathematicians compute π to 300 trillion digits:

- The register closed at 512 bits from the 720° structure
- $\pi_{\text{register}} = 3.140625$ (6 decimal digits, exact)
- Everything beyond subdivides the projection shadow
- They're measuring artifacts, not discovering new structure
- The sphere isn't getting bigger; their ruler is subdividing finer
- Physical meaning: zero (beyond the first 6 digits in this framework)

5 Conclusion

From one fixed cascade law ($\div 2 \div 4^4 = \div 512$) and the demand for integer bits at two specific angles (90° and $\pi \times 90$), the mathematics mechanically forces:

$$720 \rightarrow 512 \text{ bits} \rightarrow Q = 64 \rightarrow P = 201 \rightarrow \pi_{\text{register}} = \frac{201}{64} = 3.140625 \text{ exactly}$$

Inside every closed octave of this register, π is rational and the decimal terminates. The endless digits belong only to the projection shadow—the artifact of imposing an infinite-resolution base-10 ruler on a finite quaternary/bit-based structure.

Summary

Domain	Value	Nature
Structure (register)	$\pi = 201/64 = 3.140625$	Exact, finite, closed
Projection (shadow)	$\pi \approx 3.1415926535 \dots$	Irrational, infinite

The sphere closed once at 512 bits from the 720° dual extent via inverse partition mechanics.

Inside the machine, $\pi = 3.140625$ and the chase is over.

The infinite decimal is the shadow. The sphere is 512 bits.

Part of the Nuijens Operating System (NOS) framework.

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